MILITARY SPECIFICATION

COMPONENT BOARD ASSEMBLY: 10558242

This specification is approved for use within the US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a printed circuit component board, part of Power Supply Test Panel Subassembly MX-8051/TPM-22A, a part of Radar Test Set AN/TPM-22A.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts , Equipment and Tools for Army Materiel; Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance, and Disposition of
MIL-C-46357	Computer, Gun Directions; M18 Manufacturing Processes and Inspection of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, U S Army Armament, Munitions and Chemical Command, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publics-tions form a-part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (See 6.3)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

10558242	Component Board Assembly

PACKAGING DATA SHEET

10558242	Packaging of Component Board Assembly

(Copies of specifications, standards, handbooks, drawings, and“ publications required by the manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards) , the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3* REQUIREMENTS

3.1 Fabrication. The Component Board shall be manufactured in accordance with Drawing 10558242 and drawings pertaining thereto.

3.2 General specifications. Where applicable, the Component Board shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

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3.2.1 MIL-F-13926.

a. Dimensions and tolerances

b. Inorganic protective surface finishings

c. Part identification and marking

d. Workmanship

e. Soldering electrical connections

3.2.2 MIL-C-46357.

a. Etched circuit assemblies

b. Wiring

c. Threaded fasteners

d. Marking

e. Finishes

3.3 Environmental service conditions.

3.3.1 Vibration. The component board shall be mounted on the vibration table in such a manner that one of the vibration axes will be perpendicular to the plane of the board. The component board shall meet the requirements of 3.4 and 3.5 after being subject to continuous vibration for two hours along each of-three mutually perpendicular axes within the frequency range and amplitude as-follows:

FREQUENCY RANGE (Hz + 2 PERCENT}	EXCURSION (DOUBLE AMPLITUDE) (INCHES + 10 PERCENT)
5 to 15	0.06 inch
15 to 25	0.04 inch
25 to 55	0.02 inch

The rate of change of frequency at each frequency range shall not exceed one cycle-per-second per second (1 cps/see) . Vibration tests shall be conducted under ambient room-conditions.

3.3.2 Shock. The component board shall not suffer damage and shall meet requirements of 3.4 and 3.5 after being subjected to 18 impact shocks, consisting of three shocks in opposite directions along each of the three mutually perpendicular axes. The shock impulses shall have an acceleration of 15 gravity units (G's) and a pulse duration of 11 ± 1 milliseconds. The G value of each impulse shall be within + 10 percent when measured with a 0.2 to 250 Hz filter and the maximum G shall occur at approximately 5-1/2 milliseconds.

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3.3.3 Altitude.

3.3.3.1 Operating. The component board shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg) .

3.3.3.2 Nonoperating. The component board shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The component board shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between +32°F and +131°F.

3.3.4.2 Nonoperating (storage) . The component board shall not suffer permanent damage when subjected to the constant temperature extreme of -60°F for a period of 24 hours and +160°F for a period of 48 hours (after temperature stabilization) , and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity-nonoperating. The component board shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical.

3.4.1 Continuity. Each conductor shall be continuous electrically as specified on the schematic diagram 10558242.

3.4.2 Resistance. Resistance values shall be in accordance with Table I.

TABLE I. Resistance values.

Between Terminals		Resistance (Ohms)
E31	E36	20 ± 1%
E32	E33	20 ± 1%
E42	E44	3.6K ± 5%
E45	E44	3.6K ± 5%
E52	E57	169 ± 1%
E53	E54	43.2 ± 1%
E55	E56	140 ± 1%
E40	E37	36 ± 5%
E41	E46	36 ± 5%

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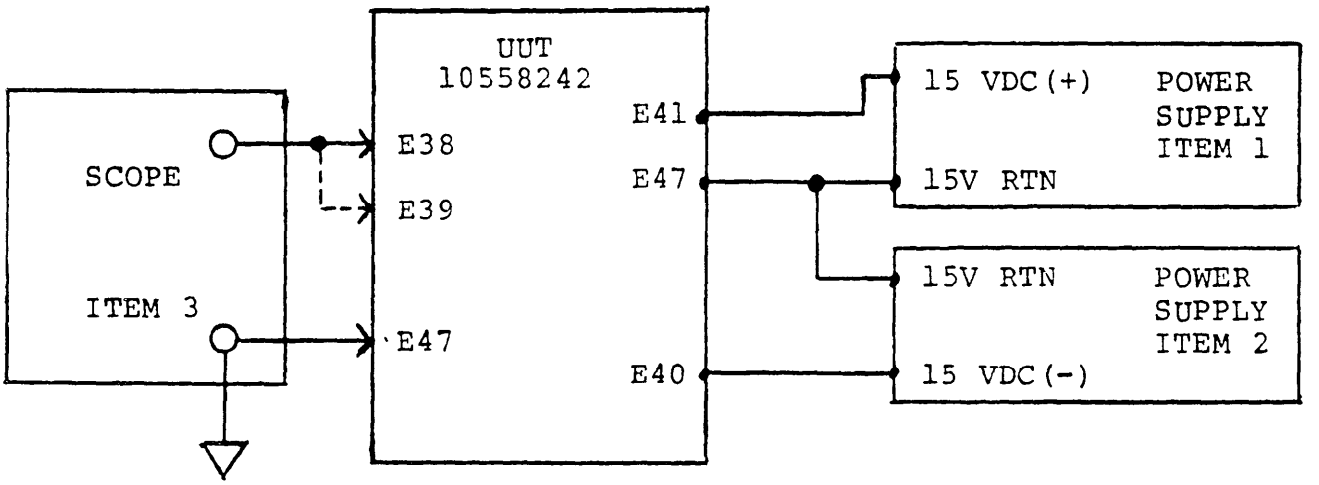


FIGURE I. Circuit connections.

3.5 Performance.

3.5.1 Conditions of operation. Unless otherwise specified, the requirements of 3.5.2 shall be met with the circuit connections shown in Figure 1.

TABLE II. Pulse voltage characteristics.

LINE	PULSE EXCURSION VDC	1/2 CYCLE PERIODS SECONDS	RISE TIME MICROSEC	FALL TIME MICROSEC
1	0.0 + l.O TO 13.0 ± 1.0	0.32 TO 0.65	0.1	0.1
2	-13.0 ± 1.0 TO 0.0 + 1.5	0.32 TO 0.65	0.1	0.1

3.5.2 Output signals.

3.5.2.1 +13v Pulse. With conditions of 3.5.1 applied and the oscilloscope connected between E38 and E47, observe a square wave in accordance with line 1 of Table II.

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3.5.2.2 -13V Pulse. With conditions of 3.5.1 applied and the oscilloscope connected between E39 and E47, observe a square wave in accordance with line 2 of Table II.

3.6 First article inspection. Unless otherwise specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2)

3.7 Workmanship. The requirements of MIL-STD-454 and other applicable specifications shall apply.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor or the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2. First article (initial production) approval. The requirement for first article approval and the responsibility (Government or contractor) for the first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three assemblies manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the

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sample to ensure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certification of conformance for materials.

4.2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certification for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product. Unless otherwise specified by the Contracting Officer, inspection lot size, lot information and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests. Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables III and IV herein, and Methods of inspection in 4.6. Examinations and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot.

TABLE 111. Classification of defects
(100 percent inspection)

	Characteristic	Requirement	Test procedure
101.	Continuity	3.4.1	4.6.2.1
102.	Resistance	3.4.2	4.6.2.2
103.	+13V Pulse	3.5.2.1	4.6.3.2
104.	-13V Pulse	3.5.2.2	4.6.3.3

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component, or test specimen in the sample to comply with any of the stated requirements.

4.3.3.1 Acceptance and rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmitance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

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4.4 Special sampling. One component board out of the first 25 produced and one out of each 25 produced thereafter shall be selected at random. The sample shall have met the requirements and tests in Table III and shall then meet the requirements and tests in Table IV.

TABLE IV. Special sampling

	Characteristic	Requirement	Test procedure
301.	Fabrication	3.1	Applicable drawings Visual
302.	General specification	3.2	Visual/Tactile
303.	Vibration	3.3.1	4.6.1.1
304.	Shock	3.3.2	4.6.1.2
305.	Altitude-operating	3.3.3.1	4.6.1.3
306.	Altitude-nonoperating	3.3.3.2	4.6.1.3
307.	Temperature	3.3.4	4.6.1.4
308.	Humidity-nonoperating	3.3.5	4.6.1.5

4.4.1 Failure of sample. Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and a resubmitted sample has been approved (see 4.3.3) .

4.5 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Contractor-furnished inspection equipment.

4.5.2.1 Contractor design. The contractor shall design and supply inspection equipment compatible with the “Methods of inspection” specified in 4.6 of this specification and with the “Component Inspection” procedures specified in “Examination” and “Test Facilities” requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus , concept construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit

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and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.2 Test equipment. In conjunction with 4.5.2.1, the test equipment found in Table V shall be used in the performance of the the applicable tests of 4.6.

TABLE V. Test equipment list

Item	Equipment	Description
1.	Power supply	15 Vdc ± 3% Regulated AC ripple: Less than 1% Capacity: 0.350A
2.	Power supply	15 Vdc ± 3%, Regulated AC ripple: Less than 1% Capacity: 0.350A
3.	Oscilloscope	Vertical Passband: dc to 10 Mhz Input Sensitivity: 1 mv per division Horizontal Deflection: 0.2 usec per division to 0.5 sec. per division
4.	Ohm Meter	To measure up to 4.0K Ohms ± 1%, continuity or infinity

4.6 Methods of inspection.

4.6.1 Environmental. Environmental tests shall be conducted in accordance with the general requirements and test methods of MIL-STD-810 except as specified herein. Testing equipment used in environmental tests shall be in accordance with “Test Facilities and Apparatus” requirements of MIL-STD-810.

4.6.1.1 Vibration. The test shall be performed as specified in Method 514, procedure 1, Category I, as modified by 3.3.1 herein. After vibration, the assembly shall be tested as given in the subparagraphs of 4.6.2 and 4.6.3 for compliance with 3.3.1.

4.6.1.2 Shock. This test shall be performed as specified in Procedure I, Method 516 of MIL-STD-810 except that shock pulses shall be 11 ± 1 ms and 15 G's. The G value of each impulse shall be within +10 percent when measured with a 0.2 to 250 Hz filter and the maximum G shall occur at approximately 5 1/2 milliseconds. After the shock impulses, the assembly shall be tested as specified in 4.6.2 and 4.6.3 for compliance with 3.3.2.

4.6.1.3 Altitude-nonoperating and operating. This test shall be performed as specified in Procedure II and III, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.6.3 for compliance with 3.3.3.

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4.6.1.4 Temperature.

4.6.1.4.1 High (operating and nonoperating) This test shall be performed as specified in Procedure I and II, "Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.6.2 and 4.6.3 for compliance with 3.5.

4.6.1.4.2 Low (operting and nonoperating) This test shall be performed as specified in Procedure I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3*3.4. The operational checkout shall be as specified in 4.6.2 and 4.6.3 for compliance with 3.5.

4.6. 1.5 Humidity-nonoperating. This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the end of the tenth cycle only, and at standard ambient conditions.

4.6.2 Electrical.

4.6.2.1 Continuity. Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.1.

4.6.2.2 Resistance. Resistance shall be measured with an ohmmeter , item 4 of Table V, for compliance with 3.4.2.

4.6.3 Performance tests.

4.6.3.1 Conditions of operation. Unless otherwise specified, all subsequent test will be performed with the D.C. power supplies and oscilloscope connected to conform to the schematic of Figure 1. Adjust the D.C. Power supplies to the values indicated.

4.6.3.2 +13V Pulse. Connect Channel 1 of the oscilloscope, item 3, of Table IV, to E38 of the component board. Check for presence of the positive pulse per 3.5.2.1.

4.6.3.3 -13V Pulse. Connect Channel 1 of the oscilloscope, item 3, of Table IV, to E39 of the component board. Check for presence of the negative pulse per 3.5.2.2.

5. PACKAGING

5.1 Packaging Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10558242. The level of protection shall be as specified in the contract.

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6. NOTES

6.1 Intended use. This component board assembly is part of the Power Supply Test Panel, Subassembly MX8051/TPM-22A which is used for troubleshooting and performance checking of specific weapon system components.

6.2 ordering data. Procurement documents should specify the following:

a. Title, number, and date of this and the item detail specifications.

b. Applicable stock number.

c. Quality Conformance Inspection, if other than specified in Section 4 of item detail specification.

d. First Article Sample requirements, if other than specified in Section 4 of item detail specification.

e. Packaging requirements, if other than specified in Section 5 of item detail specification.

f. Certificate of conformance for each lot or shipment of product.

6.3 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by and identified as US Army Armament Research and Development Command (ARRADCOM) , Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDC.

6.4 Subject term (keyword) listing.

a. Fire control (small caliber)
b. Circuit component board (electronic)
c. Military specification (Army)
d. Weapon (Vulcan weapon system M163 & M167)

Custodian:
Army-AR

Preparing activity
Army-AR

(Project 4931-A207)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-70512	COMPONENT BOARD ASSEMBLY: 10558242

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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